

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250269504

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Tsai; Yu-Ping

ENGAGING STRUCTURE FOR HAND TOOL

Abstract

An engaging structure for a hand tool includes a main body provided with a mounting element having six drive portions and six receiving portions. Each of the drive portions includes a first face, a second face, and a third face. A first intersection point is located at an intersection of the first face and the second face. A second intersection point is located at an intersection of the first face and the third face. Each of the receiving portions is an arcuate concave face. The mounting element has a first hexagon. The first face of each of the drive portions coincides with each of the sides of the first hexagon. The first hexagon has six first angled corners. The first arc of each of the six receiving portions is situated outside of each of the six first angled corners.

Inventors: Tsai; Yu-Ping (Taichung, TW)

Applicant: Tsai; Yu-Ping (Taichung, TW)

Family ID: 1000007817535

Appl. No.: 18/587688

Filed: February 26, 2024

Publication Classification

Int. Cl.: B25B23/10 (20060101); B25B13/04 (20060101); B25B13/06 (20060101); B25B13/46 (20060101)

U.S. Cl.:

CPC B25B23/105 (20130101); B25B13/04 (20130101); B25B13/065 (20130101); B25B13/462 (20130101)

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a hand tool and, more particularly, to an engaging structure for a hand tool.

Description of the Related Art

[0002] A conventional socket structure was disclosed in the U.S. Publication No. 2007/0125204, and comprises a socket **20** having an engaging hole **22**, six grooves **23**, and six stop ribs **24**.

[0003] However, such a conventional socket structure has the following disadvantages.

[0004] 1. When the socket **20** drives the screw head **40**, the stop ribs **24** are locked on the six sides of the screw head **40**, to rotate the screw head **40**. The stop ribs **24** protrude in the engaging hole **22** to an extent, and a large interval is defined between the groove **23** of the engaging hole **22** and the angled corner of the screw head **40** so that the engaging hole **22** cannot easily rub the angled corner of the screw head **40**. But, the socket **20** needs to have a determined thickness to keep the rotation force and torque, so that the outer diameter of the socket **20** is increased due to projection of the stop ribs **24**.

[0005] 2. Each of the stop ribs **24** has a sharp pointed shape so that the rotation torque is not large enough. Thus, the screw head **40** is easily worn out or chamfered by the stop ribs **24** during a period of time.

[0006] 3. When the socket **20** is used to rotate the screw head **40**, the stop ribs **24** engage the screw head **40** successively and easily cause a serration on the screw head **40**.

BRIEF SUMMARY OF THE INVENTION

[0007] In accordance with the present invention, there is provided an engaging structure for a hand tool comprising a main body. The main body is provided with a mounting element. The mounting element includes six drive portions and six receiving portions. Each of the six drive portions includes a first face, a second face, and a third face. A first intersection point is located at an intersection of the first face and the second face. A second intersection point is located at an intersection of the first face and the third face. Each of the six receiving portions is an arcuate concave face. Each of the six receiving portions is formed with a first arc. The mounting element has a first center, a first circle, and a first hexagon. The first face of each of the six drive portions coincides with each of the six sides of the first hexagon. The first center is provided with a first vertical line. The first vertical line passes through the first intersection point. The first hexagon has six first angled corners. The first arc of each of the six receiving portions is situated outside of each of the six first angled corners.

[0008] According to the primary advantages of the present invention, the main body is rotated clockwise to screw the screw member by a large area contact of the second face. Alternatively, the main body is rotated counterclockwise to unscrew the screw member by a large area contact of the third face. Alternatively, the main body is rotated counterclockwise to unscrew the worn screw member by a force concentration of the second intersection point. Thus, the mounting element has three rotational modes to drive and rotate the screw member.

[0009] According to another advantage of the present invention, each of the six receiving portions is an arcuate concave face. Each of the six receiving portions is arranged between the second face of one of the six drive portions and the third face of another one of the six drive portions. Each of the six receiving portions is formed with a first arc. The first arc is not tangent to the second face and the third face. Thus, when the main body is rotated to drive the screw member, each of the six angled corners of the screw member is received in each of the six receiving portions, thereby preventing each of the six angled corners of the screw member from being worn out.

[0010] Further benefits and advantages of the present invention will become apparent after a careful reading of the detailed description with appropriate reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0011] FIG. **1** is a perspective view of an engaging structure for a hand tool in accordance with the first preferred embodiment of the present invention.

[0012] FIG. **2** is a front view of the engaging structure for a hand tool in accordance with the preferred embodiment of the present invention.

[0013] FIG. **3** is a schematic front view of the engaging structure for a hand tool in accordance with the preferred embodiment of the present invention.

[0014] FIG. **4** is a front view of a first operation state of the engaging structure for a hand tool in accordance with the preferred embodiment of the present invention.

[0015] FIG. **5** is a front view of a second operation state of the engaging structure for a hand tool in accordance with the preferred embodiment of the present invention.

[0016] FIG. **6** is a front view of a third operation state of the engaging structure for a hand tool in accordance with the preferred embodiment of the present invention.

[0017] FIG. **7** is a perspective view of an engaging structure for a hand tool in accordance with the second preferred embodiment of the present invention.

[0018] FIG. **8** is a perspective view of an engaging structure for a hand tool in accordance with the third preferred embodiment of the present invention.

[0019] FIG. **9** is a perspective view of an engaging structure for a hand tool in accordance with the fourth preferred embodiment of the present invention.

[0020] FIG. **10** is a perspective view of an engaging structure for a hand tool in accordance with the fifth preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0021] Referring to the drawings and initially to FIGS. **1-3**, an engaging structure (or an operating structure or a driving structure) for a hand tool in accordance with the preferred embodiment of the present invention comprises a main body **10**. The main body **10** is provided with a mounting element **11**. The mounting element **11** includes six drive portions **12** and six receiving portions **13**. The six drive portions **12** and the six receiving portions **13** are arranged alternately and annularly. The main body **10** is a box end wrench or an engaging member of a hand tool.

[0022] Each of the six drive portions **12** includes a first face **121**, a second face **122**, and a third face **123**. The first face **121** is arranged between the second face **122** and the third face **123**. Each of the first face **121**, the second face **122**, and the third face **123** has a planar shape. A first intersection point **124** is located at an intersection of the first face **121** and the second face **122**. A second intersection point **125** is located at an intersection of the first face **121** and the third face **123**. A first distance **126** is defined between two parallel first faces **121**. A second distance **127** is equal to a length of the first face **121**. The second distance **127** is defined between the first intersection point **124** and the second intersection point **125**. The second distance **127** is one twentieth of the first distance **126**.

[0023] Each of the six receiving portions **13** is an arcuate concave face. Each of the six receiving portions **13** is arranged between the second face **122** of one of the six drive portions **12** and the third face **123** of another one of the six drive portions **12**. Each of the six receiving portions **13** is formed with a first arc **131**. The first arc **131** is not tangent to the second face **122**. The first arc **131** is not tangent to the third face **123**. The first arc **131** has a first radius **132** which is one tenth of the first distance **126** and is two times of the second distance **127**.

[0024] In a two-dimensional view, the mounting element **11** has an axis provided with a first center **14** which constructs a first circle **141**. The first face **121** of each of the six drive portions **12** is tangent to the first circle **141**. The first circle **141** forms a first hexagon **142** having six sides. The first face **121** of each of the six drive portions **12** coincides with each of the six sides of the first

hexagon **142**. The first center **14** is provided with a first vertical line **143**. The first vertical line **143** passes through the first intersection point **124**. The first face **121** and the third face **123** are located at a first side of the first vertical line **143**, and the second face **122** is located at a second side of the first vertical line **143**.

[0025] The first hexagon **142** has six first angled corners **144**. The first arc **131** of each of the six receiving portions **13** is situated outside of each of the six first angled corners **144**. A first angle **145** is defined between each of the six sides of the first hexagon **142** and the second face **122**. The first angle **145** is ranged between four degrees and five degrees) (4° - 5°). A second angle **146** is defined between each of the six sides of the first hexagon **142** and the third face **123**. The second angle **146** is more than the first angle **145**. The value of the second angle **146** minus that of the first angle **145** is ranged between 0.5 and 1.5. Preferably, the value of the second angle **146** minus that of the first angle **145** is ranged between 1.0 and 1.5.

[0026] Referring to FIG. 4 with reference to FIGS. 1-3, the main body **10** is used to operate and rotate a screw member **20** having six sides and six angled corners. When the main body **10** is rotated clockwise to drive the screw member **20**, the second face **122** of each of the six drive portions **12** rests on each of the six sides of the screw member **20**, a space (or an interval) is defined between the third face **123** of each of the six drive portions **12** and each of the six sides of the screw member **20**, and each of the six angled corners of the screw member **20** is received in each of the six receiving portions **13**. Thus, when the main body **10** is rotated clockwise relative to the screw member **20**, each of the six receiving portions **13** is spaced from each of the six angled corners of the screw member **20**, to prevent each of the six angled corners of the screw member **20** from being worn.

[0027] Referring to FIG. 5 with reference to FIGS. 1-3, when the main body **10** is rotated counterclockwise to drive the screw member **20**, the third face **123** of each of the six drive portions **12** rests on each of the six sides of the screw member **20**, a space (or an interval) is defined between the second face **122** of each of the six drive portions **12** and each of the six sides of the screw member **20**, and each of the six angled corners of the screw member **20** is received in each of the six receiving portions **13**. Thus, when the main body **10** is rotated counterclockwise relative to the screw member **20**, each of the six receiving portions **13** is spaced from each of the six angled corners of the screw member **20**, to prevent each of the six angled corners of the screw member **20** from being worn.

[0028] Referring to FIG. 6 with reference to FIGS. 1-3, the screw member **20** is worn out and rounded. When the main body **10** is rotated counterclockwise to drive the screw member **20**, the third face **123** and the second intersection point **125** of each of the six drive portions **12** rest on each of the six sides of the screw member **20**. In such a manner, the second intersection point **125** has a linear contact with and concentrates a force on the screw member **20** to drive the screw member **20**. Thus, the mounting element **11** is used to drive and rotate the screw member **20** that is worn.

[0029] Referring to FIG. 7, the main body **10** is a ratchet wheel structure pivotally mounted on a ratchet wrench.

[0030] Referring to FIG. 8, the main body **10** is a socket structure.

[0031] Referring to FIG. 9, the main body **10** is a sliding rod structure.

[0032] Referring to FIG. 10, the main body **10** is a float connector or a universal connector or a T-shaped universal connector.

[0033] In another preferred embodiment of the present invention, a fillet (or round corner) is formed between each of the six receiving portions **13** and the second face **122** of each of the six drive portions **12**, so that each of the six receiving portions **13** and the second face **122** of each of the six drive portions **12** are connected smoothly. A fillet (or round corner) is formed between each of the six receiving portions **13** and the third face **123** of each of the six drive portions **12**, so that each of the six receiving portions **13** and the third face **123** of each of the six drive portions **12** are

connected smoothly. The second distance **127** is one tenth of the first distance **126**. Thus, the first radius **132** is equal to the second distance **127**.

[0034] Accordingly, the engaging structure for a hand tool of the present invention has the following advantages.

[0035] 1. With reference to FIG. 4, when the main body **10** is rotated clockwise to drive the screw member **20**, the second face **122** of each of the six drive portions **12** provides a primary force, and the first face **121** of each of the six drive portions **12** provides a secondary force. Thus, the second face **122** has the maximum contact area with the screw member **20**, to prevent the six sides of the screw member **20** from being worn out.

[0036] 2. With reference to FIG. 5, when the main body **10** is rotated counterclockwise to drive the screw member **20**, the third face **123** of each of the six drive portions **12** provides a primary force, and the first face **121** of each of the six drive portions **12** provides a secondary force. Thus, the third face **123** has a larger contact area with the screw member **20**, to prevent the six sides of the screw member **20** from being worn out.

[0037] 3. With reference to FIG. 6, when the screw member **20** is worn out and rounded, the second intersection point **125** of each of the six drive portions **12** is locked on each of the six sides of the screw member **20**. Thus, the second intersection point **125** of each of the six drive portions **12** applies and focuses a force on the screw member **20** to effectively drive and unscrew the screw member **20** that is worn and rounded.

[0038] 4. The main body **10** is rotated clockwise to screw the screw member **20** by a large area contact of the second face **122**. Alternatively, the main body **10** is rotated counterclockwise to unscrew the screw member **20** by a large area contact of the third face **123**. Alternatively, the main body **10** is rotated counterclockwise to unscrew the worn screw member **20** by a force concentration of the second intersection point **125**. Thus, the mounting element **11** has three rotational modes to drive and rotate the screw member **20**.

[0039] 5. Each of the six receiving portions **13** is an arcuate concave face. Each of the six receiving portions **13** is arranged between the second face **122** of one of the six drive portions **12** and the third face **123** of another one of the six drive portions **12**. Each of the six receiving portions **13** is formed with a first arc **131**. The first arc **131** is not tangent to the second face **122** and the third face **123**. Thus, when the main body **10** is rotated to drive the screw member **20**, each of the six angled corners of the screw member **20** is received in each of the six receiving portions **13**, thereby preventing each of the six angled corners of the screw member **20** from being worn out.

[0040] 6. Each of the first face **121**, the second face **122**, and the third face **123** has a planar shape, so that the first face **121**, the second face **122**, and the third face **123** will not cause a bite mark (or serration) on the screw member **20** during rotation.

[0041] 7. In practice, the mounting element **11** includes more than twenty metric specifications and more than twenty British specifications, so that it will take too much time to design the sizes and specifications of the mounting element **11**. In the engaging structure for a hand tool of the present invention, the second distance **127** is one twentieth of the first distance **126**, and the first radius **132** is one tenth of the first distance **126**. The mathematic expressions of the mounting element **11** are built in a software. Thus, after the value of the first distance **126** is designed and finished, the values of the second distance **127** and the first radius **132** are also determined, without having to design the variables of each of the specifications of the mounting element **11**, thereby greatly decreasing the cost of fabrication.

[0042] Although the invention has been explained in relation to its preferred embodiment(s) as mentioned above, it is to be understood that many other possible modifications and variations can be made without departing from the scope of the present invention. It is, therefore, contemplated that the appended claim or claims will cover such modifications and variations that fall within the scope of the invention.

Claims

1. A engaging structure for a hand tool, comprising: a main body; wherein: the main body is provided with a mounting element; the mounting element includes six drive portions and six receiving portions; the six drive portions and the six receiving portions are arranged alternately and annularly; each of the six drive portions includes a first face, a second face, and a third face; the first face is arranged between the second face and the third face; each of the first face, the second face, and the third face has a planar shape; a first intersection point is located at an intersection of the first face and the second face; a second intersection point is located at an intersection of the first face and the third face; a first distance is defined between two parallel first faces; a second distance is equal to a length of the first face; the second distance is defined between the first intersection point and the second intersection point; the second distance is one twentieth of the first distance; each of the six receiving portions is an arcuate concave face; each of the six receiving portions is arranged between the second face of one of the six drive portions and the third face of another one of the six drive portions; each of the six receiving portions is formed with a first arc; the first arc is not tangent to the second face; the first arc is not tangent to the third face; the first arc has a first radius which is one tenth of the first distance and is two times of the second distance; the mounting element has an axis provided with a first center which constructs a first circle; the first face of each of the six drive portions is tangent to the first circle; the first circle forms a first hexagon having six sides; the first face of each of the six drive portions coincides with each of the six sides of the first hexagon; the first center is provided with a first vertical line; the first vertical line passes through the first intersection point; the first face and the third face are located at a first side of the first vertical line, and the second face is located at a second side of the first vertical line; the first hexagon has six first angled corners; the first arc of each of the six receiving portions is situated outside of each of the six first angled corners; a first angle is defined between each of the six sides of the first hexagon and the second face; the first angle is ranged between four degrees and five degrees; a second angle is defined between each of the six sides of the first hexagon and the third face; the second angle is more than the first angle; the value of the second angle minus that of the first angle is ranged between 0.5 and 1.5; the value of the second angle minus that of the first angle is ranged between 1.0 and 1.5; the main body is used to rotate a screw member having six sides and six angled corners; when the main body is rotated clockwise to drive the screw member, the second face of each of the six drive portions rests on each of the six sides of the screw member, a space is defined between the third face of each of the six drive portions and each of the six sides of the screw member, and each of the six angled corners of the screw member is received in each of the six receiving portions; when the main body is rotated clockwise relative to the screw member, each of the six receiving portions is spaced from each of the six angled corners of the screw member, to prevent each of the six angled corners of the screw member from being worn; when the main body is rotated counterclockwise to drive the screw member, the third face of each of the six drive portions rests on each of the six sides of the screw member, a space is defined between the second face of each of the six drive portions and each of the six sides of the screw member, and each of the six angled corners of the screw member is received in each of the six receiving portions; when the main body is rotated counterclockwise relative to the screw member, each of the six receiving portions is spaced from each of the six angled corners of the screw member, to prevent each of the six angled corners of the screw member from being worn; when the screw member is worn out and rounded and when the main body is rotated counterclockwise to drive the screw member, the third face and the second intersection point of each of the six drive portions rest on each of the six sides of the screw member; the second intersection point has a linear contact with and concentrates a force on the screw member to drive the screw member; and the mounting element is used to drive and rotate the screw member that is worn.

2. The engaging structure for a hand tool as claimed in claim 1, wherein the main body is a box end wrench or an engaging member of a hand tool.
 3. The engaging structure for a hand tool as claimed in claim 1, wherein the main body is a ratchet wheel structure or a socket structure.
 4. The engaging structure for a hand tool as claimed in claim 1, wherein the main body is a sliding rod structure, a float connector, a universal connector or a T-shaped universal connector.
 5. The engaging structure for a hand tool as claimed in claim 1, wherein: a fillet is formed between each of the six receiving portions and the second face of each of the six drive portions, so that each of the six receiving portions and the second face of each of the six drive portions are connected smoothly; and a fillet is formed between each of the six receiving portions and the third face of each of the six drive portions, so that each of the six receiving portions and the third face of each of the six drive portions are connected smoothly.
 6. The engaging structure for a hand tool as claimed in claim 1, wherein the second distance is one tenth of the first distance, and the first radius is equal to the second distance.
-